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Problem Set 2.

1). Let $A_{n \times n}$ be a matrix such that

$BA = I$ and $B'A = I$

for some $B_{n \times n}$ and $B'_{n \times n}$

Now, Let A be such that

$Ax = y$ for some x and y.

Hence B is $By = x$ for that x and y.

$\therefore B(y) = x$

$B(Ax) = x$

$\boxed{BA = I}$ — B is left inverse of A

Now, Let C be such that

$Cy = x$ for that x and y.

So, $A(x) = y$

$A(Cy) = y$

$\boxed{AC = \cancel{y}.I.}$ C is right inverse of

$\therefore$ As left and right inverse BOTH exist for A, A is an invertible matrix where $|A| \neq 0$.

Now, Let $C = A^{-1}$ be inverse of A

where $AA^{-1} = I$

Hence, as

$$BA = B'A = I$$

$$BAA^{-1} = B'AA^{-1}$$

$$BI = B'I$$

$$\boxed{B = B'}$$

Hence, No, It's NOT possible for ~~there~~ to be two distinct matrices where

$BA = B'A = I$ and $B \neq B'$.

2). Let A, X, Y be ~~a~~ matrices such that

$$AX = Y$$

So, Augmented matrix A is such that

$$A' = [A|Y].$$

Now, let A be an invertible matrix. Hence let A^{-1} be such that

$$AA^{-1} = I. \text{ and } A^{-1}A = I$$

So,
$$AX = Y$$
$$A^{-1}AX = A^{-1}Y$$
$$X = A^{-1}Y$$

So, for $AA^{-1} = I.$

$Y = I$ and $A^{-1} = X$

So, ~~X~~ ~~solution~~

$$A' = [A|I],$$

But, ~~as~~ for $A^{-1}A = I.$

$$(A^{-1})' = [A^{-1}|I]$$

So, ~~AX~~ now reducing $[A|I]$

gives $[I \mid A^{-1}]$.

<u>example</u> :

for matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$,

$$\underbrace{\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}}_{A}^{2\times 2} \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_{X}^{2\times 1} = \underbrace{\begin{bmatrix} y_1 \\ y_2 \end{bmatrix}}_{Y}^{2\times 1}$$

$$\therefore A' = \left[\begin{array}{cc|c} 1 & 2 & y_1 \\ 3 & 4 & y_2 \end{array}\right].$$

Now,

for $AA^{-1} = I$,

$$A' = \left[\begin{array}{c|c} A & I \end{array}\right]$$

$$= \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 4 & 0 & 1 \end{array}\right].$$

performing operations to make it rref:

$R_2 \leftarrow R_2 - 3R_1$

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & -2 & -3 & 1 \end{array}\right]$$

$R_1 \leftarrow R_1 + R_2$

$$\left[\begin{array}{cc|cc} 1 & 0 & -2 & 1 \\ 0 & -2 & -3 & 1 \end{array}\right]$$

$R_2 \leftarrow \frac{R_2}{-2}$

$$\left[\begin{array}{cc|cc} 1 & 0 & -2 & 1 \\ 0 & 1 & \frac{3}{2} & \frac{-1}{2} \end{array}\right].$$

$$\left[\begin{array}{c|c} I & A^{-1} \end{array}\right].$$

$$\therefore A^{-1} = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & \frac{-1}{2} \end{bmatrix}.$$

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Verification

$$AA^{-1} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & \frac{-1}{2} \end{bmatrix}$$

$$= \begin{bmatrix} -2+3 & 1-1 \\ -6+6 & 3-2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= I.$$

3). Let A be a matrix

$$A = \begin{bmatrix} 0 & 5 & -1 \\ 1 & 8 & -1 \\ 2 & 11 & 1 \end{bmatrix} \quad X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}.$$

where $AX = Y$. $\quad Y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}.$

Step 1: Let A' be augmented matrix such that ~~$AX = Y$~~

$$A' = [A \mid Y]$$

$2y_2 + 3y_1$

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$$\therefore A' = \left[\begin{array}{ccc|c} 0 & 5 & -1 & y_1 \\ 1 & 8 & -1 & y_2 \\ 2 & 1 & 1 & y_3 \end{array}\right].$$

Step 2: Now we now reduce A such that

$$[A \mid Y] \xrightarrow{rref} [R \mid Z]$$

where R is rref of A.

Step 3: Now, we need to make sure there are non-zero rows in Z corresponding to zero rows in R.

So:

$R_2 \leftrightarrow R_1$

$$A' = \left[\begin{array}{ccc|c} 1 & 8 & -1 & y_2 \\ 0 & 5 & -1 & y_1 \\ 2 & 1 & 1 & y_3 \end{array}\right].$$

$R_2 \leftarrow \frac{R_2}{5}$

$\frac{8}{5} \cdot \frac{-5}{5} - \frac{5+8}{5}$

$5 - \frac{6}{5}$

$$\left[\begin{array}{ccc|c} 1 & 8 & -1 & y_2 \\ 0 & 1 & \frac{-1}{5} & \frac{y_1}{5} \\ 2 & 1 & 1 & y_3 \end{array}\right]$$

$R_1 \leftarrow R_1 - 8R_2$

$$\left[\begin{array}{ccc|c} 1 & 0 & \frac{3}{5} & y_2 - \frac{8y_1}{5} \\ 0 & 1 & \frac{-1}{5} & \frac{y_1}{5} \\ 2 & 1 & 1 & y_3 \end{array}\right]$$

$R_3 \leftarrow R_3 - 2R_1$

$$\left[\begin{array}{ccc|c} 1 & 0 & \frac{3}{5} & y_2 - \frac{8y_1}{5} \\ 0 & 1 & \frac{-1}{5} & \frac{y_1}{5} \\ 0 & 1 & \frac{-1}{5} & y_3 - 2y_2 + \frac{16y_1}{5} \end{array}\right]$$

$R_3 \leftarrow R_3 - R_2$

$$\left[\begin{array}{ccc|c} 1 & 0 & \frac{3}{5} & y_2 - \frac{8y_1}{5} \\ 0 & 1 & \frac{-1}{5} & \frac{y_1}{5} \\ 0 & 0 & 0 & y_3 - 2y_2 + 3y_1 \end{array}\right]$$

Now, ~~where~~ if

$$y_3 - 2y_2 + 3y_1 \neq 0,$$

then it's an inconsistent system.

However, if

$$y_3 - 2y_2 + 3y_1 = 0.$$

Hence

$$\begin{bmatrix} 1 & 0 & \frac{8}{5} \\ 0 & 1 & \frac{-1}{5} \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} y_2 - \frac{8y_1}{5} \\ \frac{y_1}{5} \\ y_3 - 2y_2 + 3y_1 \end{bmatrix}$$

$$\therefore \quad x_1 + \frac{3x_3}{5} = y_2 - \frac{8y_1}{5}$$

$$x_2 - \frac{x_3}{5} = \frac{y_1}{5}$$

$$y_3 - 2y_2 + 3y_1 = 0$$

$\therefore \boxed{x_2 = \frac{y_1 + x_3}{5}}$

Let $x_3 = a$

$\therefore x_1 + \frac{3a}{5} = \frac{5y_2 - 8y_1}{5}$

$\boxed{x_1 = \frac{5y_2 - 8y_1 - 3a}{5}}$

$\therefore$ solution set of X

$\boxed{= \left(\frac{5y_2 - 8y_1 - 3a}{5}, \frac{y_1 + a}{5}, a \right).}$

$a \in R$

provided that

$\boxed{y_3 - 2y_2 + 3y_1 = 0.}$

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4). Given: $A_{n\times n}$.

To prove:

(i) (A is invertible) $\iff$ ($AX = Y$ has a solution for each $Y_{n\times 1}$) (ii).

Proof:

I). forward proof:

Assume A is an invertible matrix.

$\therefore \exists A^{-1}$ such that

$$AA^{-1} = A^{-1}A = I.$$

Now, as

$AX = Y$ for some $X_{n\times 1}$, $Y_{n\times 1}$,

$$A^{-1}AX = A^{-1}Y$$
$$IX = A^{-1}Y$$
$$X = A^{-1}Y.$$

So, for every Y,

$A^{-1}Y = X$ has some X as a solution set.

Now, if $A^{-1}Y = X_1$ and

$A^{-1}Y = X_2$,

then $X_1 = X_2$

$\therefore$ The solution is unique.

So, $AX = Y$ has a solution $\not\exists$ for each $Y_{n \times 1}$.

II) **backward proof.**

Assume $AX = Y$ has a solution for each column matrix $Y_{n\times 1}$.

~~So, Let $Y = I$~~

So, $A' = [A \mid Y]$

which is Augmented matrix of A'.

Let R be RREF of A

So. $A' = \left[\begin{array}{c|c} R & Z \end{array} \right]$

Now, $Z = \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{bmatrix}$.

Now, if R has r zero rows,

then z_n must also have r zero rows

as otherwise $AX = Y$ will be an inconsistent system (contradiction)

But, Y ~~doesn't have~~ must have a solution for each Y.

So, there must be a y where $z_n, z_{n-1}, \ldots, z_{n-r}$ are NON ZERO.

$\therefore$ R CANNOT have any ~~rest~~ zero rows.

Now, as RREF of A (R) has NO zero rows, the ~~system must be consistent~~.

This means there is a leading non zero entry for each column.

thus $R = I_n$.

So,

$A \sim I_n$,

which means A is invertible.

$\therefore$ as (i) $\Rightarrow$ (ii)

and (ii) $\Rightarrow$ (i)

(i) $\Longleftrightarrow$ (ii).

Hence, proved.

5). Let $A_{1\,n\times n}$, ~~such~~ $A_{2\,n\times n}$ - - - $A_{k\,n\times n}$
$A_{n\times n}$,
such that

$$A = A_1 A_2 \cdots A_k.$$

Given : A is invertible.

To prove : $A_1, A_2, \ldots, A_k$ are also invertible.

Now, A is invertible.

$\therefore \exists A^{-1}$ (unique)
such that

$$AA^{-1} = A^{-1}A = I.$$

Also, $(B_1 B_2 \ldots B_k)^{-1} = B_k^{-1} B_{k-1}^{-1} \cdots B_1^{-1}$.

Now, using Inductive proof:

~~**~~ if $\exists B$ such that $AB = I$
then ~~and~~ $\exists C$ such that $CA = I$
~~then AC~~ where $C = B$
and A is invertible.

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Base case:

$k = 2$.

$\therefore A = A_1 A_2$

so, $AA^{-1} = I$

$\therefore AA^{-1} = (A_1 A_2)(A)^{-1}$

$I = (A_1 A_2)(A)^{-1}$

$I = A_1(A_2 A^{-1})$

so, $A_1 \cdot (A_2 A^{-1}) = I$

$\therefore \boxed{A_1^{-1} = A_2 A^{-1}}$

$\therefore A_1$ is invertible.

Using strong induction,

Inductive hypothesis:

Assume it's true for $k = \ell - 1$.

$\therefore (A_1 \ldots \ldots A_{\ell-1})(A^{-1}) = I$

$(A_{\ell-1})^{-1} = (A \ldots \ldots A_{\ell-2}) A^{-1}$.

and A_{e-1} is invertible

~~for~~

$\therefore A_1, A_2 --- A_{e-1}$ are all invertible matrices.

~~So, for A~~

Inductive step:

for A_e,

$\cancel{A_e (A_{e-1} \cdot A_{e-2} \cdots A_1) \cdot A_e^{-1} = A}$.

~~Now,~~

$\cancel{(A_e)(A_{e-1} A_{e-2} \cdots A_1) A_e (A_1)^{-1} (A_2)^{-1}} \cdots (A_{e-1})^{-1}$

$= A(A_1^{-1} A_2^{-1} -- (A_{e-1})^{-1})$.

$\cancel{A_e =} A(A_1^{-1} A_2^{-1} \cdots A_{e-1})^{-1}$

~~So,~~

for A_e,

$A = A_1 A_2 \cdots A_{e-1} A_e$.

Now,

$I = A^{-1}(A_1 A_2 \cdots A_{e-1}) \times (A_e)$

so,

$(A_e)^{-1} = A^{-1} A_1 A_2 \cdots A_{e-1}$.

$\therefore$ there exists an inverse for A_e.

Hence proved,

for invertible $A = A_1 A_2 \cdots A_k$,

A_i is invertible $\forall i \in [1, k]$.

6). A is invertible $n \times n$

<u>To prove</u>:

If a sequence of elementary row operations reduce A to I, then the same operations on I lead to A^{-1}.

<u>Proof</u>:

<u>Theorem used</u>: If A is $n \times n$ matrix,

A is invertible $\Leftrightarrow A \sim I_n \Leftrightarrow A = E_1 E_2 E_3 \cdots E_k$

where $E_1, E_2 \cdots E_k$ are elementary row operations.

So, if A is invertible, from theorem, it can be row reduced to I_n using k elementary row operations.

$\because e_i(I) = E_i$,

$$\boxed{I \cancel{A} = E_1 E_2 \cdots E_k A}$$

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Now, $AA^{-1} = A^{-1}A = I$,

Multiplying with A^{-1},

$$IA^{-1} = (E_1 E_2 \cdots E_k) AA^{-1}$$

$$\boxed{A^{-1} = (E_1 E_2 \cdots E_k) I}$$

So, performing the SAME sequence of operations to I yields A^{-1}.

Hence proved.

7) Given : A has a left inverse.

∴ for some B,

$$BA = I,$$

To prove : $AX = 0$ has only a trivial solution.

Proof : $AX = 0$

$$\therefore B(AX) = B(0)$$
$$(BA)X = 0$$
$$IX = 0$$
$$\boxed{X = 0}$$

So, the only solution for X_n is that

$$x_1 = x_2 = \dots x_n = 0.$$

So, $AX = 0$ has only a trivial solution $X = 0$ if A has a left inverse,

8) Given : $A_{m \times n}$ over field F.

$u_1, u_2 \dots u_r$ are non-zero row vectors in the rref of A.

Sol^n:

Properties of v_i, $i \in [1, r]$.

i). 1st non-zero entry is 1 in column k_i
ii). other rows have entries in k_ith column as 0
iii). $k_i \neq k_j$ for any i, j $(i \neq j) \in [1, r]$.
so, no two rows can have leading entry in the same column.

So,

for v_1, $A_{1k_1} = 1$
But $A_{ik_1} = 0$ for $i \in [2, r]$

Using Inductive Proof:

Base case: $r = 1$

$\therefore c_1 v_1 = 0$.

Now, as U_1 is Not a zero row,

$U_1 \neq 0$

$\therefore \; c_1 U_1 = 0$

only if $c_1 = 0$.

Inductive Hypothesis:

assume for some k,

$$\sum_{i=1}^{k} c_i v_i = 0.$$

so, $c_1 v_1 + c_2 v_2 + \ldots + c_k v_k = 0$

$\therefore \; c_1 = c_2 = c_3 = \ldots = c_k = 0$

Inductive Step:

for some $k+1$,

$$\sum_{i=1}^{k+1} c_i v_i = 0.$$

so, $\underbrace{c_1 v_1 + \cdots c_k v_k}_{=0 \text{ from}} + c_{k+1} v_{k+1} = 0.$

Inductive Hypothesis.

$\therefore c_{k+1} v_{k+1} = 0$

$\because v_{k+1} \neq 0$

$c_{k+1} = 0.$

Hence proved, It's TRUE.

For scalars $c_1, \cdots c_r \in F$,

if $\sum_{i=1}^{r} c_i v_i = 0$, then $c_i = 0$ for all $i \in \{1, 2, \cdots r\}$,